

LOCAL CACCIOPPOLI AND ROUGH-CORE EXCLUSION

ABSTRACT. We present a fully explicit local Caccioppoli estimate for renormalized Navier–Stokes fields. All terms produced by the localized energy inequality are estimated separately, and all absorptions are written explicitly. The argument uses standard PDE notation and the usual integrability classes of suitable weak solutions.

In particular, we derive local windowed H^1 control on compact space–time cylinders from local compact Caffarelli–Kohn–Nirenberg quantities. The result gives the rough-core alternative as the negation of local compact-CKN control.

This paper is written in the classical framework of suitable weak solutions (Leray [Ler34], Caffarelli–Kohn–Nirenberg [CKN82], Scheffer [Sch76], Lin [Lin98]).

1. LOCALIZED EQUATION AND ENERGY INEQUALITY

Let (V, P, a, b) be defined on the compact renormalized cylinder

$$Q_{2R,I} = B_{2R} \times I,$$

where $I \subset \mathbb{R}$ is a bounded interval and $B_{2R} \subset \mathbb{R}^3$ is the ball of radius $2R$. Assume

$$(1) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V = a(\tau)V + a(\tau)(y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0,$$

with viscosity constant $\nu > 0$.

Throughout, for a space–time set $E \subset \mathbb{R}^3 \times \mathbb{R}$, let

$$\int_E f := \int_E f(y, \tau) dy d\tau,$$

for $r > 0$ and a time interval $J \subset \mathbb{R}$, define

$$Q_{r,J} := B_r \times J, \quad |Q_{r,J}| := |B_r| |J|.$$

and for a ball $B \subset \mathbb{R}^3$, denote the spatial mean of pressure by

$$(P)_B(\tau) = \frac{1}{|B|} \int_B P(y, \tau) dy.$$

A suitable weak solution on $Q_{2R,I}$ satisfies the local energy inequality: for every nonnegative $\phi \in C_c^\infty(Q_{2R,I})$ and every $\tau_1 < \tau_2$ in I ,

$$(2) \quad \int_{B_{2R}} \frac{|V(\cdot, \tau_2)|^2}{2} \phi(\cdot, \tau_2) - \int_{B_{2R}} \frac{|V(\cdot, \tau_1)|^2}{2} \phi(\cdot, \tau_1) + \nu \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \phi |\nabla V|^2 \leq \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} (\partial_\tau \phi + \Delta \phi) + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \left(\frac{|V|}{2} \right. \\ \left. + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} (aV + a(y \cdot \nabla V) + b \cdot \nabla V) \cdot V \right) \phi.$$

2. COMPACT-CYLINDER CACCIOPPOLI ESTIMATE

Theorem 2.1 (Compact-cylinder Caccioppoli estimate). *Assume that*

$$\nabla \cdot V = 0, V \in L_t^\infty L_x^2(Q_{2R,I}) \cap L_t^2 H_x^1(Q_{2R,I}), P \in L_t^{3/2} L_x^{3/2}(Q_{2R,I}),$$

and

$$a, b \in L^\infty(I), \int_{Q_{2R,I}} |V|^3 < \infty, \quad \int_{Q_{2R,I}} |P - (P)_{B_{2R}}(\tau)|^{3/2} < \infty.$$

Let $I' \Subset I$ be compact. Then there is a constant

$$C = C(R, \nu, \|a\|_{L^\infty(I)}, \|b\|_{L^\infty(I)}, \text{dist}(I', \partial I))$$

such that

$$\mathcal{H}_{I'}(R) = \int_{I'} \int_{B_R} |\nabla V|^2 \leq C \left(1 + \int_I \int_{B_{2R}} |V|^3 + \int_I \int_{B_{2R}} |P - (P)_{B_{2R}}(\tau)|^{3/2} \right).$$

Proof. Step 1: localized cut-off and support localization. Set

$$\delta := \text{dist}(I', \partial I) > 0.$$

Choose $\eta \in C_c^\infty(I)$ and $\zeta \in C_c^\infty(B_{2R})$ with

$$0 \leq \eta \leq 1, \quad 0 \leq \zeta \leq 1, \quad \eta \equiv 1 \text{ on } I', \quad \zeta \equiv 1 \text{ on } B_R,$$

and

$$\|\eta'\|_{L^\infty(I)} \leq C_\eta \delta^{-1}, \quad \|\nabla \zeta\|_{L^\infty(B_{2R})} \leq C_\zeta R^{-1}, \quad \|\Delta \zeta\|_{L^\infty(B_{2R})} \leq C_\zeta R^{-2}.$$

Set

$$\phi(y, \tau) := \eta(\tau) \zeta(y)^2.$$

Then $\phi \in C_c^\infty(Q_{2R, I})$, $0 \leq \phi \leq 1$, and

$$\phi \equiv 1 \text{ on } B_R \times I'.$$

Choose $\tau_1 < \tau_2$ in I such that $\eta(\tau_1) = \eta(\tau_2) = 0$ and $I' \subset (\tau_1, \tau_2) \subset I$. Insert this test function into (2). The terms at τ_1, τ_2 vanish, so

$$\nu \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \phi |\nabla V|^2 \leq |R_\tau| + |R_\Delta| + |R_{\text{conv}}| + |R_{\text{pres}}| + |R_{\text{mod}}|,$$

where

$$\begin{aligned} R_\tau &:= \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} \partial_\tau \phi, & R_\Delta &:= \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} \Delta \phi, \\ R_{\text{conv}} &:= \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \left(\frac{|V|^2}{2} V \right) \cdot \nabla \phi, & R_{\text{pres}} &:= \int_{\tau_1}^{\tau_2} \int_{B_{2R}} (P - (P)_{B_{2R}}(\tau)) V \cdot \nabla \phi, \\ R_{\text{mod}} &:= \int_{\tau_1}^{\tau_2} \int_{B_{2R}} (aV + a(y \cdot \nabla V) + b \cdot \nabla V) \cdot V \phi. \end{aligned}$$

Define

$$A := \int_{Q_{2R, I}} |V|^2, B := \int_{Q_{2R, I}} |V|^3, D := \int_{Q_{2R, I}} |P - (P)_{B_{2R}}(\tau)|^{3/2}, X := \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \phi |\nabla V|^2.$$

By Hölder on $Q_{2R, I}$,

$$A \leq |Q_{2R, I}|^{1/3} B^{2/3} \leq C_{R, I} (1 + B).$$

Since $\phi(y, \tau) = \eta(\tau) \zeta(y)^2$,

$$\partial_\tau \phi = \eta'(\tau) \zeta(y)^2, \quad \nabla \phi = 2\eta(\tau) \zeta(y) \nabla \zeta(y), \quad \Delta \phi = \eta(\tau) \Delta(\zeta(y)^2),$$

hence

$$\|\partial_\tau \phi\|_{L^\infty(Q_{2R, I})} \leq C_\eta \delta^{-1}, \quad \|\nabla \phi\|_{L^\infty(Q_{2R, I})} \leq 2C_\zeta R^{-1}, \quad \|\Delta \phi\|_{L^\infty(Q_{2R, I})} \leq C_\zeta R^{-2}.$$

Step 2: derivative terms from the cut-off. Hence

$$|R_\tau| \leq \frac{1}{2} \|\partial_\tau \phi\|_{L^\infty(Q_{2R, I})} A \leq \frac{1}{2} C_\eta \delta^{-1} A,$$

$$|R_\Delta| \leq \frac{1}{2} \|\Delta \phi\|_{L^\infty(Q_{2R, I})} A \leq \frac{1}{2} C_\zeta R^{-2} A.$$

Hence

$$|R_\tau| + |R_\Delta| \leq C_0(\delta^{-1} + R^{-2})(1 + B).$$

Step 3: convection term. Because $\nabla\phi \in L^\infty(Q_{2R,I})$,

$$|R_{\text{conv}}| \leq \|\nabla\phi\|_{L^\infty(Q_{2R,I})} \int_{Q_{2R,I}} |V|^3 \leq C_\nabla B.$$

Step 4: pressure term. For each time τ ,

$$\int_{B_{2R}} (P - (P)_{B_{2R}}(\tau)) V \cdot \nabla\phi = \int_{B_{2R}} PV \cdot \nabla\phi,$$

with

$$\int_{B_{2R}} (P)_{B_{2R}}(\tau) V \cdot \nabla\phi = (P)_{B_{2R}}(\tau) \int_{B_{2R}} V \cdot \nabla\phi = -(P)_{B_{2R}}(\tau) \int_{B_{2R}} \phi \nabla \cdot V = 0$$

(using $\phi \in C_c^\infty(B_{2R})$ and $\nabla \cdot V = 0$). Hence

$$|R_{\text{pres}}| \leq \|\nabla\phi\|_{L^\infty(Q_{2R,I})} \int_{\tau_1}^{\tau_2} \|P - (P)_{B_{2R}}(\tau)\|_{L_x^{3/2}(B_{2R})} \|V(\tau, \cdot)\|_{L_x^3(B_{2R})} d\tau \leq \|\nabla\phi\|_{L^\infty(Q_{2R,I})} \|P - (P)_{B_{2R}}\|_{L_t^{3/2} L_x^{3/2}(Q)}$$

Since

$$\|P - (P)_{B_{2R}}\|_{L_t^{3/2} L_x^{3/2}(Q_{2R,I})} = D^{2/3}, \quad \|V\|_{L_t^3 L_x^3(Q_{2R,I})} = B^{1/3},$$

with $x = D^{2/3}$ and $y = B^{1/3}$, Young's inequality $xy \leq \frac{2}{3}x^{3/2} + \frac{1}{3}y^3$ yields

$$|R_{\text{pres}}| \leq C_\phi D^{2/3} B^{1/3} \leq C_\phi(B + D).$$

Step 5: modulation terms. Split

$$R_{\text{mod}} = R_a + R_{ay} + R_b,$$

where

$$R_a := \int_{\tau_1}^{\tau_2} \int_{B_{2R}} a \phi |V|^2, \quad R_{ay} := \int_{\tau_1}^{\tau_2} \int_{B_{2R}} a \phi (y \cdot \nabla V) \cdot V, \quad R_b := \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \phi (b \cdot \nabla V) \cdot V.$$

For R_a ,

$$|R_a| \leq \|a\|_{L^\infty(I)} A \leq C(1 + B).$$

For R_{ay} , since $|y| \leq 2R$ on $\text{supp } \phi$,

$$|R_{ay}| \leq \|a\|_{L^\infty(I)} \left(\int_{Q_{2R,I}} \phi |\nabla V|^2 \right)^{1/2} \left(\int_{Q_{2R,I}} \phi |y|^2 |V|^2 \right)^{1/2} \leq 2R \|a\|_{L^\infty(I)} \left(\int_{Q_{2R,I}} \phi |\nabla V|^2 \right)^{1/2} A^{1/2}.$$

Using $\phi \leq 1$, we also have

$$\left(\int_{Q_{2R,I}} \phi |y|^2 |V|^2 \right)^{1/2} \leq 2R A^{1/2},$$

so

$$|R_{ay}| \leq \|a\|_{L^\infty(I)} \left(\int_{Q_{2R,I}} \phi |\nabla V|^2 \right)^{1/2} \cdot 2R A^{1/2}.$$

Applying Young's inequality in the form

$$uv \leq \frac{\nu}{8} u^2 + \frac{1}{\nu} v^2, \quad u = \left(\int_{Q_{2R,I}} \phi |\nabla V|^2 \right)^{1/2}, \quad v = 2R \|a\|_{L^\infty(I)} A^{1/2},$$

gives

$$|R_{ay}| \leq \frac{\nu}{8} \int_{Q_{2R,I}} \phi |\nabla V|^2 + \frac{4R^2 \|a\|_{L^\infty(I)}^2}{\nu} A.$$

Therefore

$$|R_{ay}| \leq \frac{\nu}{8} \int_{Q_{2R,I}} \phi |\nabla V|^2 + C(1+B).$$

For R_b ,

$$|R_b| \leq \|b\|_{L^\infty(I)} \left(\int_{Q_{2R,I}} \phi |\nabla V|^2 \right)^{1/2} \left(\int_{Q_{2R,I}} \phi |V|^2 \right)^{1/2} \leq \frac{\nu}{8} \int_{Q_{2R,I}} \phi |\nabla V|^2 + \frac{\|b\|_{L^\infty(I)}^2}{\nu} A \leq \frac{\nu}{8} \int_{Q_{2R,I}} \phi |\nabla V|^2 + C(1+B)$$

Step 6: absorption and conclusion. Collecting estimates from Steps 2–5 yields

$$\nu X \leq C_0(1+B) + C_1B + C_2D + \frac{\nu}{4}X,$$

for constants C_0, C_1, C_2 depending on

$$R, \nu, \|a\|_{L^\infty(I)}, \|b\|_{L^\infty(I)}, \delta.$$

Subtract the last term from the left-hand side:

$$\frac{3\nu}{4}X \leq C(1+B+D),$$

where $C = \frac{4}{3\nu}(C_0 + C_1)$. Since $\phi \equiv 1$ on $B_R \times I'$ and $I' \subset (\tau_1, \tau_2)$,

$$\mathcal{H}_{I'}(R) \leq X \leq C \left(1 + \int_{I'} \int_{B_{2R}} |V|^3 + \int_{I'} \int_{B_{2R}} |P - (P)_{B_{2R}}(\tau)|^{3/2} \right),$$

which is the claimed estimate after renaming the constant. $\square$

Definition 2.2 (Local Caffarelli–Kohn–Nirenberg control on compact cylinders). For $r > 0$ and compact $J \subset I$, define

$$\mathcal{C}_J(r) := \int_J \int_{B_r} |V|^3, \quad \mathcal{D}_J(r) := \int_J \int_{B_r} |P - (P)_{B_r}(\tau)|^{3/2}, \quad \mathcal{H}_J(r) := \int_J \int_{B_r} |\nabla V|^2.$$

We say that local Caffarelli–Kohn–Nirenberg control holds on $J \times B_r$ if both $\mathcal{C}_J(r)$ and $\mathcal{D}_J(r)$ are finite, and local windowed H^1 control holds if $\mathcal{H}_J(r) < \infty$.

Corollary 2.3 (Rough-core alternative). Assume $a, b \in L^\infty(I)$ and let $R > 0$, $I' \Subset I$.

- (i) If local Caffarelli–Kohn–Nirenberg control holds on $I \times B_{2R}$, then for every compact $I' \Subset I$ there exists

$$C = C(R, I', \nu, \|a\|_{L^\infty(I)}, \|b\|_{L^\infty(I)}, \mathcal{C}_I(2R), \mathcal{D}_I(2R))$$

such that

$$\mathcal{H}_{I'}(R) = \int_{I'} \int_{B_R} |\nabla V|^2 \leq C.$$

- (ii) If local windowed H^1 control fails in every compact inner cylinder, then there exists at least one compact inner cylinder on which local Caffarelli–Kohn–Nirenberg control fails. Equivalently,

$$[\forall R' \in (0, R], \forall J \Subset I : \mathcal{H}_J(R') = +\infty] \implies [\exists R'' \in (0, R], \exists J' \Subset I : \mathcal{C}_{J'}(R'') = +\infty \text{ or } \mathcal{D}_{J'}(R'') = +\infty].$$

Proof. (i) If local Caffarelli–Kohn–Nirenberg control holds on $I \times B_{2R}$, then

$$\mathcal{C}_I(2R) < \infty, \quad \mathcal{D}_I(2R) < \infty.$$

Applying Theorem 2.1 gives

$$\mathcal{H}_{I'}(R) \leq C,$$

hence (i) follows.

(ii) By contraposition, it suffices to show that if the Caffarelli–Kohn–Nirenberg quantities are finite on every compact inner cylinder, then local windowed H^1 control is finite on every compact inner cylinder.

Assume

$$\forall R' \in (0, R], \forall J \in I : \mathcal{C}_J(R') < \infty \text{ and } \mathcal{D}_J(R') < \infty.$$

For any fixed pair (R', J) , applying Theorem 2.1 at scale R' and interval J gives

$$\mathcal{H}_J(R') < \infty.$$

Because (R', J) was arbitrary, this gives local windowed H^1 control on every compact inner cylinder, which is precisely the negation of the premise in (ii). $\square$

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